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Quantisation of Conductance in Single-Atom Junctions

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Abstract

Metallic contacts that are reduced to only a few atoms exhibit transport properties that no longer follow classical circuit theory. In such constrictions, the flow of electrons is governed by quantum mechanics, and the electrical conductance appears in distinct steps of the conductance quantum $G_0 = 2e^2/h$. This experiment investigates this behaviour using a mechanically controlled break junction (MCBJ), where a notched gold wire is elongated with sub-nanometre precision. As the contact narrows down to a single atom, stable conductance plateaus emerge, corresponding to the discrete number of available transmission channels.

The report outlines the theoretical foundation of quantized conductance, including the Landauer approach, the role of eigenchannels, and the limitations of the single-electron picture. Experimentally, the preparation of the wire, the tunnelling-based calibration of the piezo displacement, and the acquisition of conductance traces and histograms are discussed. From the data, the conductance quantum is extracted and compared with theoretical expectations. Additional investigations examine the effect of applied bias, the behaviour at higher conductance values, and differences observed for other materials such as copper and platinum.

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1 Introduction

1.1 Motivation

When electrical contacts are reduced to only a few atoms in width, their behaviour can no longer be described by familiar macroscopic laws. At these dimensions, the motion of electrons is governed by quantum mechanics rather than by classical continuum models. As a result, the conductance of a narrow metallic constriction changes in discrete steps instead of continuously, a phenomenon that stands in sharp contrast to the behaviour of ordinary wires or bulk materials.

Understanding transport at this scale is not only of academic interest. The ongoing miniaturisation of electronic components, the development of molecular switches, and the design of atomic-scale sensors all rely on knowing how electrons move when a device approaches the size of its constituent atoms. Single-atom contacts, for instance, provide one of the simplest settings in which the wave nature of electrons becomes directly observable. They form an ideal test bed for exploring concepts such as transmission channels, quantum interference, and ballistic transport.

The mechanically controlled break junction (MCBJ) method offers a practical way to access this regime. By bending a substrate with a notched metal wire, one can create and repeatedly adjust a contact that narrows down to a single atom. This level of control makes it possible to record reproducible conductance traces and analyse the characteristic steps that emerge as the junction is opened and closed. The experiment provides a hands-on opportunity to study these effects using a table-top setup, making quantum transport accessible without the need for low temperatures or ultra-high vacuum conditions.

1.2 Aim of the Experiment

The primary goal of this experiment is to observe and analyse the quantisation of electrical conductance in atomic-scale metallic contacts. By repeatedly breaking and reforming a gold junction with sub-nanometre precision, the conductance can be recorded as a function of the

electrode separation. The resulting traces are expected to show stable plateaus close to integer multiples of the conductance quantum

$$G_0 = \frac{2e^2}{h},$$

which reflects the number of transmission channels available for electron transport through the narrowest part of the contact.

A second objective is to become familiar with the MCBJ technique and the calibration procedures required to operate it reliably. This includes preparing the notched wire, establishing the tunnelling regime used for converting piezo voltage into electrode displacement, and identifying appropriate settings for piezo speed, measurement rate, and bias voltage.

Finally, the experiment aims to deepen the understanding of the Landauer approach to quantum transport, in which the conductance is determined by the transmission probabilities of a small number of electronic modes. By comparing theoretical expectations with experimentally obtained conductance traces and histograms, the validity and limitations of this picture can be explored in a quantitative way.

2 Theoretical Background

2.1 Ohm's Law and Classical Conductance

In macroscopic conductors, the relation between current I and applied voltage U is described by Ohm's law,

$$I = \frac{U}{R},$$

where the resistance R depends on material properties and geometry. For a uniform conductor,

$$R = \rho \frac{L}{A},$$

with ρ denoting the resistivity, L the length of the wire, and A its cross-sectional area. This classical description assumes that electrons move diffusively, undergoing frequent scattering with impurities, lattice vibrations, or structural imperfections.

When a metallic contact is reduced to atomic dimensions, these assumptions fail. The electron mean free path can exceed the size of the constriction, meaning that carriers no longer scatter inside the contact region. Transport then becomes ballistic, and the wave nature of electrons must be taken into account. In this regime, the classical resistance formula no longer applies, and a quantum-mechanical description is required [4, 5].

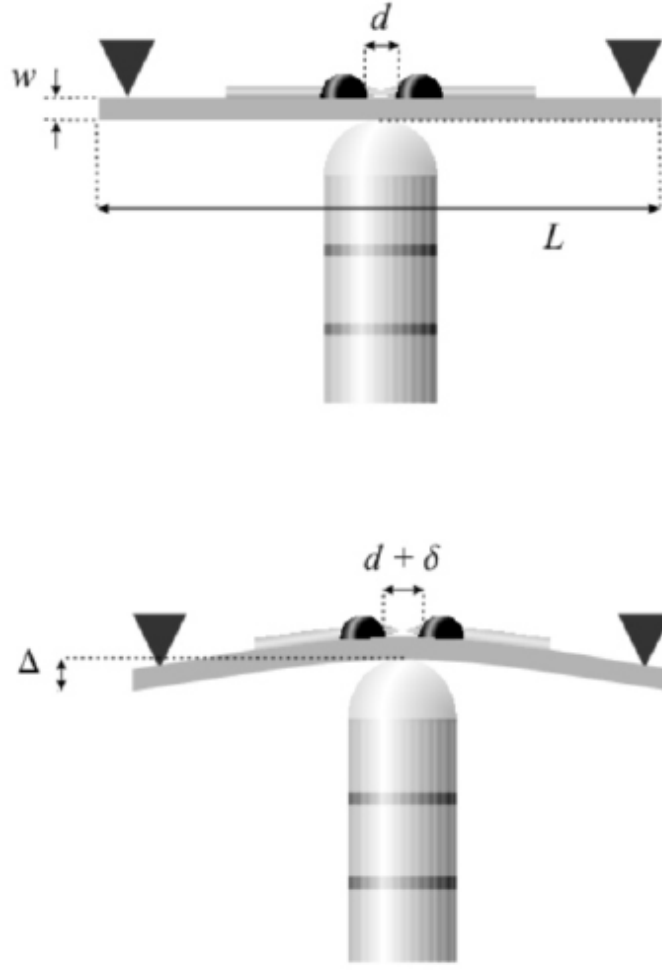


Figure 2.1: Conceptual comparison between diffusive electron transport in macroscopic conductors and ballistic transport in atomic-scale contacts [7].

2.2 Quantum Tunnelling

As two metal electrodes are separated, the electronic wavefunctions overlap less, and a potential barrier forms between them. Although an electron may not have enough classical energy to cross this barrier, the quantum wavefunction extends into the barrier region and decays exponentially. This produces a finite probability for the electron to appear on the far side, giving rise to tunnelling.

For a simple rectangular barrier of height Φ , the transmission probability can be written as

$$T \propto \exp(-2\kappa\delta), \quad \kappa = \sqrt{\frac{2m_e\Phi}{\hbar^2}},$$

where δ is the separation between the electrodes. Because this dependence is exponential, the measured tunnelling current is extremely sensitive to small variations in distance. This feature is used experimentally to calibrate the piezo-controlled displacement in the mechanically controlled break-junction setup [4, 5].

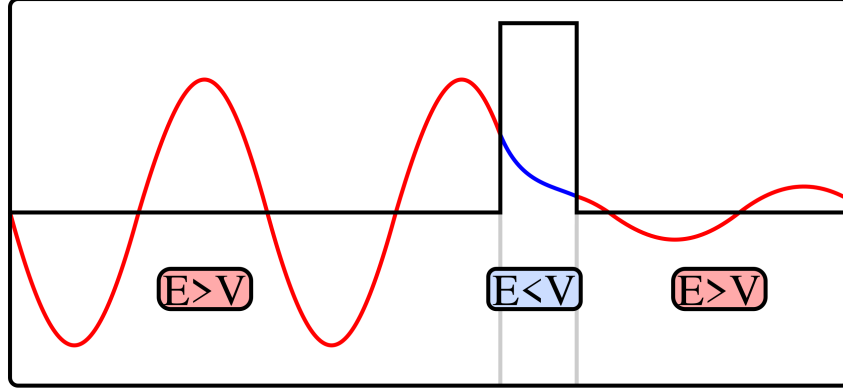


Figure 2.2: Schematic of an electron tunnelling through a potential barrier, illustrating exponential decay of the wavefunction in the classically forbidden region [11].

2.3 Fermi Wavelength and Electronic States in Metals

In a metal, conduction electrons occupy quantum states up to a maximum energy known as the Fermi energy E_F . The corresponding Fermi wavevector k_F is defined through the dispersion relation of free electrons,

$$E = \frac{\hbar^2 k^2}{2m_e}, \quad k_F = \sqrt{\frac{2m_e E_F}{\hbar^2}}.$$

The associated Fermi wavelength is

$$\lambda_F = \frac{2\pi}{k_F},$$

and is typically on the order of a few ångströms for common metals. This length scale becomes crucial when the cross-section of a conductor approaches atomic dimensions. If the width of the contact is comparable to, or smaller than, λ_F , only a limited number of transverse electron modes can fit through the constriction, and the transport properties become quantised [4, 5, 7].

In addition, only electrons with energies very close to E_F contribute significantly to electrical transport. States well below the Fermi level are mostly filled and cannot participate because the Pauli principle prevents additional electrons from occupying them. Likewise, higher-energy states are largely unoccupied at room temperature since $k_B T \ll E_F$. This restricted energy window around E_F effectively determines the number of modes that can transmit electrons through an atomic-scale contact [6, 3].

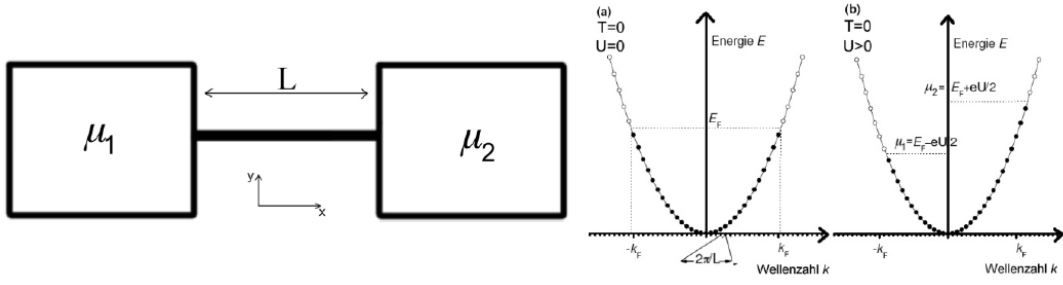


Figure 2.3: Schematic representation of electronic states in a metal, showing the occupied states up to the Fermi energy and the definition of the Fermi wavevector and wavelength [7].

2.4 Quantisation of Conductance

When a metallic contact is thinned down to a few atoms, its lateral dimensions become comparable to the Fermi wavelength of the electrons. In this regime, electrons no longer behave like a continuous fluid, but instead occupy a discrete set of transverse modes that can propagate through the constriction. Each of these modes acts like an independent channel for charge transport. A useful way to visualise this is to think of the contact as a short one-dimensional waveguide in which only certain standing-wave patterns fit.

For a single mode that is fully open, the current carried at a small bias voltage U is

$$I = \frac{2e^2}{h} U,$$

where the factor of two accounts for the two possible spin orientations. This expression leads directly to the conductance quantum,

$$G_0 = \frac{2e^2}{h},$$

which represents the maximum contribution from one perfectly transmitting channel. Since multiple modes can, in principle, propagate through the constriction, the total conductance is given by

$$G = NG_0,$$

where N counts the number of transmitting modes. These contributions are additive as long as the modes do not mix.

A key consequence of this structure is that the conductance does not change smoothly as the geometry of the contact evolves. Instead, it exhibits a sequence of plateaus at integer multiples of G_0 [10, 8, 1]. Physically, these plateaus correspond to situations in which the set of active

modes remains fixed. The atomic arrangement in the contact may shift slightly, yet the number of allowed transverse modes stays constant, and so the conductance remains nearly unchanged.

A sudden change in conductance occurs only when the constriction becomes narrow enough that one of the modes can no longer be supported, or conversely, when stretching the contact broadens it sufficiently for a new mode to appear. These transitions manifest as sharp jumps between neighbouring plateaus. The resulting step-like curves, observed when the contact is slowly opened or closed, form the characteristic signature of quantised conductance.

Although this ideal picture assumes perfectly transmitting channels, real atomic contacts can exhibit partial transmission due to scattering or slight asymmetries in the geometry. In such cases, the conductance steps may not align perfectly with integer multiples of G_0 , but the overall plateau structure remains easily recognisable in experiments. This robustness is one of the reasons why mechanically controlled break junctions provide such a clear demonstration of quantum transport at room temperature.

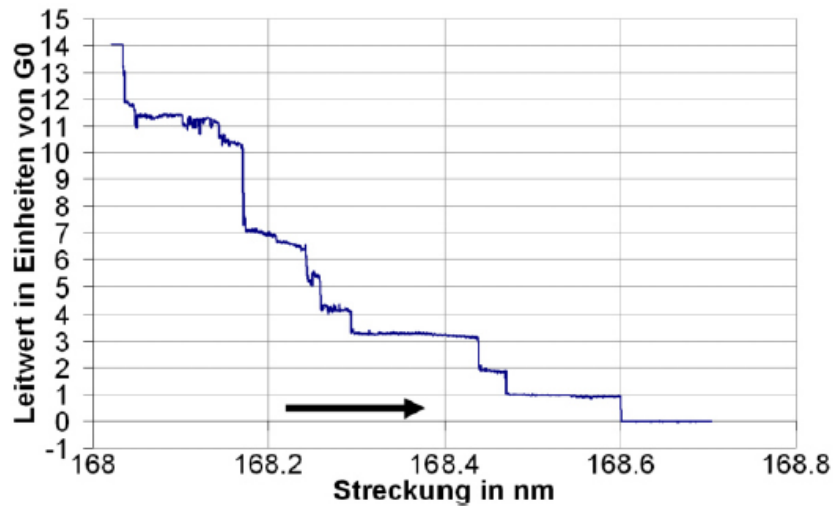


Figure 2.4: Schematic conductance trace showing the step-like behaviour characteristic of a one-dimensional constriction. Each plateau corresponds to a discrete mode contributing a quantum of conductance [7].

2.5 Landauer Formalism

When a metallic contact is reduced to the scale of only a few atoms, the familiar classical concepts of resistance and scattering lose much of their meaning. Electrons traverse such a narrow constriction ballistically, and their behaviour is governed by wave mechanics rather than by diffusive transport. The Landauer formalism provides a powerful way to describe conduction

in this regime by linking the electrical conductance to the quantum-mechanical transmission of electrons through the narrowest point of the junction [3, 1].

In this picture, the atomic contact acts as a scattering region placed between two large electron reservoirs. Electrons incident from either side can be reflected or transmitted depending on how their wavefunctions interfere in the constriction. This process is encoded in the scattering matrix

$$S = \begin{pmatrix} r & t' \\ t & r' \end{pmatrix},$$

where t describes transmission from left to right and r describes reflection back into the left reservoir. Because electrons obey coherent quantum mechanics on this length scale, the phases of the scattering amplitudes and interference between different paths play a crucial role.

The total transmission through the contact is obtained by analysing the Hermitian matrix $t^\dagger t$, whose eigenvalues

$$\tau_i, \quad 0 \leq \tau_i \leq 1,$$

represent the transmission probabilities of distinct eigenchannels [2, 9]. Each eigenchannel behaves like an independent one-dimensional conductor. Even if many transverse modes exist in the leads, the narrowest part of the junction restricts how many can fit, so only a handful of channels—often just one for gold—carry current through the contact.

A fully open channel with $\tau_i = 1$ contributes a conductance of

$$G_0 = \frac{2e^2}{h},$$

while partially open channels contribute a fraction of this value. Summing over all channels gives the Landauer formula

$$G = \frac{2e^2}{h} \sum_i \tau_i.$$

This connects the macroscopic observable (conductance) directly to quantum transmission probabilities, providing a microscopic explanation for why conductance in atomic-scale contacts appears in discrete steps. Whenever a channel opens or closes as the geometry changes, the conductance jumps by approximately one quantum.

In reality, atomic contacts seldom behave like perfect, lossless waveguides. Finite temperature produces a broadened energy distribution of electrons, which smooths the steps in the conductance curve. Inelastic processes such as phonon emission, atomic rearrangements within the junction, or weak disorder may further reduce the transparency of the channels, leading to intermediate values $0 < \tau_i < 1$. Imperfect coupling to the bulk electrodes or asymmetries in the atomic structure can also shift or distort the expected conductance values [8, 10]. Despite these

complications, the Landauer formalism remains remarkably successful and forms the theoretical backbone for interpreting measurements from mechanically controlled break junctions.

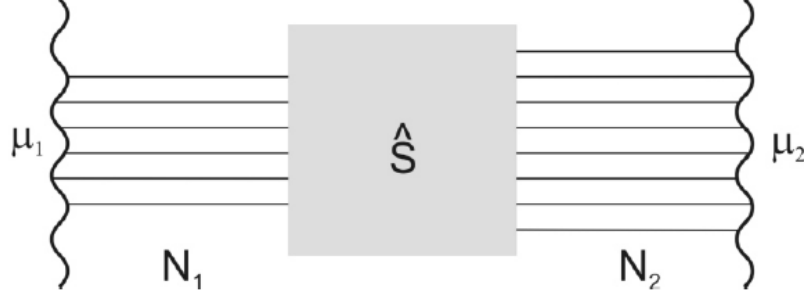


Figure 2.5: Illustration of electron transport in the Landauer picture: incoming and outgoing waves, scattering at the atomic constriction, and the decomposition into independent eigenchannels [7].

2.6 Mechanically Controlled Break Junctions (MCBJ)

The mechanically controlled break junction (MCBJ) technique provides a highly reliable and reproducible way to create metallic contacts that are only a few atoms wide. The central idea is to control the separation of a notched metal wire with sub-nanometre precision by mechanically bending the substrate on which it is mounted. This approach enables the formation, stretching, and repeated reconstruction of atomic-scale junctions in a stable and well-controlled environment [8, 1].

Three-Point Bending Geometry

The MCBJ setup is based on a simple but effective three-point bending geometry. A flexible substrate, often made of phosphor bronze or a similar spring material, is supported at two outer points while a central push rod applies force from below. When the substrate is bent upward, the top surface is put under tensile strain and any structure attached to it is correspondingly elongated.

The metallic wire, which is fixed to the substrate using two clamps or metal plates, experiences a highly amplified elongation at the position of the notch. Because the substrate bends smoothly, its displacement translates into a controlled stretching of the wire. A piezoelectric actuator mounted beneath the push rod provides extremely fine tuning of the bending curvature, allowing the electrode separation to be adjusted with sub-ångström resolution [4, 5].

The mechanical reduction ratio, defined as the factor by which piezo displacement is translated into electrode separation, is typically very large, often on the order of 10^{-5} to 10^{-6} . This gives the experimenter a tremendous degree of control over the junction.

Notched Wire and Controlled Breaking

To localise the breaking point, a shallow notch is cut into the centre of the metallic wire before it is mounted. The notch reduces the cross-sectional area at that point, ensuring that when the substrate is bent sufficiently, the wire fractures precisely where intended. This controlled breaking process is crucial, as it prevents uncontrolled deformation elsewhere along the wire.

Once the wire breaks, the two freshly created electrode tips remain very close to each other. Because the fracture typically occurs in a crystalline region of the metal, the resulting electrode surfaces are remarkably clean and atomically sharp. Small adjustments of the piezo voltage then allow the separation between the electrodes to be increased or decreased by ångström-scale increments.

As the gap widens, the system transitions smoothly from metallic contact to atomic-scale constrictions and eventually into the tunnelling regime. Closing the junction reverses this progression, allowing many cycles of breaking and reformation to be performed with high reproducibility [7].

Ballistic Regime and Atomic Rearrangements

When the constriction formed between the two electrode tips is reduced to just a few atoms, the electron mean free path far exceeds the length of the contact. As a result, electrons travel through the junction ballistically, without experiencing scattering inside the narrow region. The transport properties of the contact are therefore determined entirely by its atomic configuration and the quantum-mechanical transmission probabilities of the available modes [3, 1].

The atomic structure of the constriction is not static. As the electrodes are pulled apart, the atoms in the narrowest region rearrange themselves in response to the changing mechanical forces. These rearrangements, which can involve the formation or breaking of single-atom chains, produce sudden jumps in conductance. Between these jumps, the conductance often remains nearly constant, forming the characteristic plateaus associated with quantised transport.

Because the geometry at the atomic scale changes only discretely, the conductance also exhibits discrete transitions. This interplay between mechanical stability, atomic organisation, and quantum transmission makes the MCBJ method uniquely suited for studying conductance quantisation at room temperature. The repeated ability to form nearly identical atomic contacts

is one of the primary advantages of this technique over other methods such as STM-based break junctions.

2.7 Gear Ratio Calculation

A key feature of the mechanically controlled break junction (MCBJ) technique is the strong mechanical reduction between the motion of the piezoelectric actuator and the resulting change in electrode separation. This reduction factor, commonly referred to as the *gear ratio*, allows micrometre-scale piezo displacements to be converted into atomic-scale changes in the contact geometry. A reliable estimate of this factor is therefore essential for interpreting conductance plateaus and tunnelling measurements [4, 5].

Geometric Model of the Break Junction

To estimate the mechanical gear ratio, a simplified geometric model of the bending mechanism is employed, as described in the instruction manual. The model considers one half of the break junction, where the piezo actuator applies a vertical displacement at a point located a distance l from the center of the bending beam. The outer support points of the substrate are separated by a distance L .

Due to this lever geometry, the vertical displacement Δy imposed by the piezo is amplified at the position of the metallic wire. The effective vertical displacement at the wire position is given by

$$\Delta y' = \frac{L}{l} \Delta y, \quad (2.1)$$

where L denotes the distance between the outer supports.

As the substrate bends, this vertical displacement results in a horizontal separation of the electrode tips. Approximating the deformation geometrically as a right-angled triangle and noting that the stretching occurs symmetrically on both sides of the junction, the total horizontal displacement can be written as

$$\Delta x = 2 \left(\sqrt{(\Delta y')^2 + L^2} - L \right) = 2 \left(\sqrt{\left(\frac{L}{l} \Delta y \right)^2 + L^2} - L \right). \quad (2.2)$$

Mechanical Gear Ratio

The mechanical gear ratio is defined as the ratio between the horizontal electrode separation and the applied vertical displacement,

$$r = \frac{\Delta x}{\Delta y}. \quad (2.3)$$

Using the geometrical parameters specified in the instruction manual, namely $l = 13$ mm and $L = 14$ mm, the gear ratio was calculated as a function of the vertical displacement Δy . The resulting dependence is shown in Fig. 2.6

For the maximum vertical displacement $\Delta y_{\max} = 1$ mm, a maximum gear ratio of

$$r_{\max} \approx 0.083 \quad (2.4)$$

is obtained.

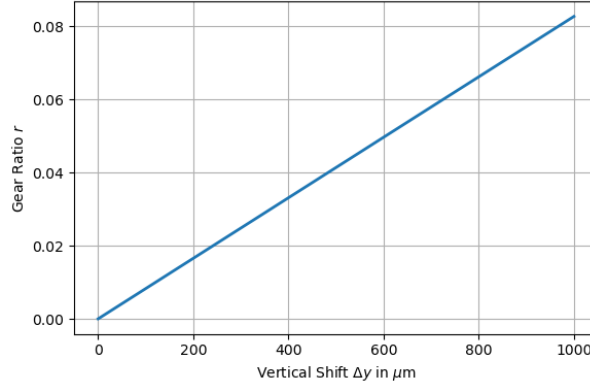


Figure 2.6: Calculated gear ratio for the setup

It should be noted that the dominant source of uncertainty in this estimation arises from the determination of the vertical displacement Δy . In the experiment, the coarse positioning of the junction is performed manually, while the piezo actuator is only used for fine adjustments once the contact is within a few micrometres of the breaking point. This manual pre-alignment inevitably introduces uncertainties, which directly propagate into the effective horizontal displacement.

3 Experimental Setup

3.1 Overview of the MCBJ Apparatus

The mechanically controlled break junction apparatus is designed to create and repeatedly adjust metallic contacts at the atomic scale with high mechanical stability. The setup combines a flexible bending substrate, precision mechanical control, and sensitive electrical measurement electronics to monitor the conductance as the contact is stretched or relaxed. Figure 3.1 summarises the essential components of the system [4, 5].

At the centre of the arrangement is a notched gold wire, which forms the actual junction. The wire is fixed to a bending beam that acts as a mechanical lever. When the substrate is deflected by the piezoelectric actuator, the notched region experiences a strongly amplified elongation, eventually forming a constriction only a few atoms wide. This geometry provides the sub-nanometre control necessary to observe quantised conductance.

To record the electrical response of the junction, the contact is connected to a low-noise transimpedance amplifier (IV converter). This amplifier converts the extremely small currents flowing through the atomic contact into measurable voltage signals. The output of the converter is routed to a data acquisition (DAQ) unit, which digitises the signal with sufficient resolution and sampling speed for real-time analysis.

The measurement software provides the interface between the experimenter and the apparatus. It displays the conductance as a function of piezo displacement, logs measurement data, and coordinates the timing between piezo actuation and signal acquisition. In automated runs, the software can generate long sequences of opening and closing traces, compute histograms, and store each individual conductance curve.

Together, these components form a highly versatile setup capable of resolving the electronic behaviour of atomic-scale junctions with exceptional precision. The combination of mechanical stability, controlled motion, and sensitive readout makes the MCBJ technique one of the most reliable methods for studying quantised conductance in metals [7, 1].

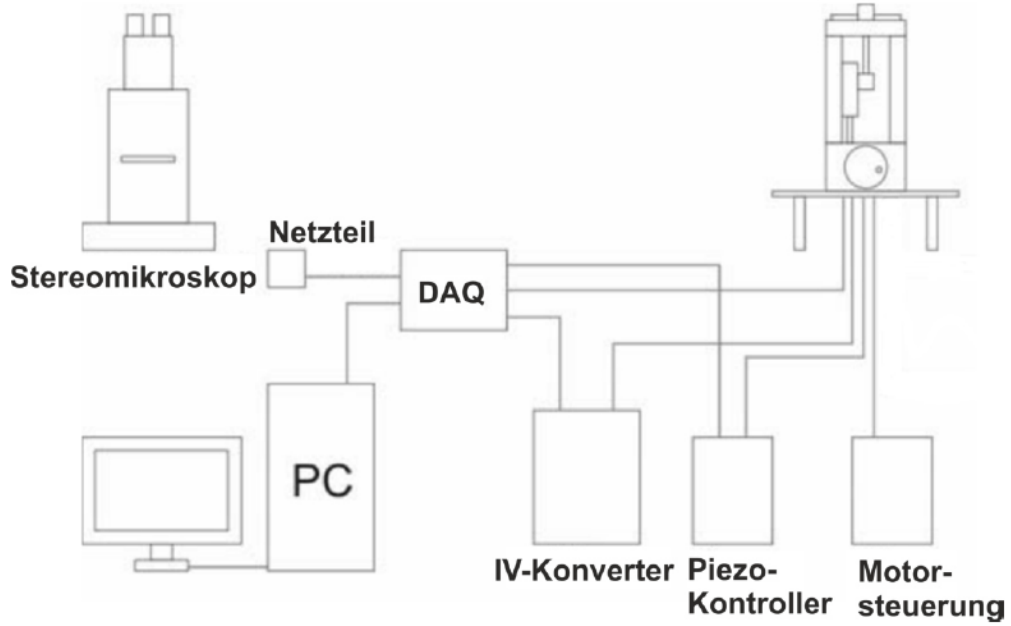


Figure 3.1: Schematic overview of the mechanically controlled break junction setup, showing the bending beam, notched wire, piezo actuator, IV converter, DAQ system, and measurement interface [7].

3.2 Sample Preparation

The preparation of the metallic wire is a crucial step in the mechanically controlled break junction experiment, as the geometry of the notch and the quality of the mounting directly influence the stability and reproducibility of the atomic contact. Gold is typically used because of its chemical inertness, high ductility, and well-understood electronic structure. In this experiment, a wire of approximately $100\text{ }\mu\text{m}$ diameter and a length of about 2 cm serves as the basis for forming the junction [4, 5].

To define the point at which the wire will break, a small notch is carefully engraved at its midpoint. This notch typically removes roughly one third of the original diameter, reducing the cross-sectional area sufficiently to ensure that the fracture occurs precisely at this location. The incision is made using a guided scalpel under a microscope, allowing controlled removal of material while keeping the remaining section mechanically stable. A clean and well-defined notch is essential for producing sharp electrode tips after breaking and for ensuring predictable deformation as the substrate is bent [7].

After engraving, the wire is mounted between two small brass plates that act as electrical contacts. The plates clamp the wire securely without deforming it and provide a low-resistance

connection to the measurement electronics. Care is taken to align the wire so that the notched region lies exactly above the bending point of the substrate. This alignment guarantees that the mechanical stress during bending is concentrated at the notch and that the wire separates cleanly when the substrate is deflected.

Once mounted, the assembly is fixed to the bending beam using screws or adhesive, depending on the specific design. The mounting must be firm enough to prevent slippage but gentle enough to avoid introducing unwanted strain into the wire. After installation, the entire assembly is inspected under a microscope to verify the notch geometry, the alignment of the wire, and the electrical integrity of the connections [1, 3].

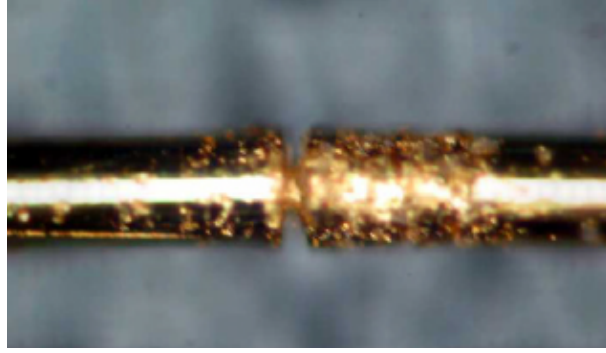


Figure 3.2: Illustration of the sample preparation process [7].

3.3 Piezo Configuration and Electrical Circuit

Precise control of the electrode separation in a break junction relies on the piezoelectric actuator mounted beneath the bending beam. The piezo element converts an applied voltage into a small mechanical displacement, allowing the junction to be opened and closed with sub-ångström precision. Its voltage range typically spans several tens of volts, within which the displacement is approximately linear. This controlled motion is essential for exploring both the metallic contact regime and the tunnelling regime that follows as the electrodes are drawn apart [4, 5].

The electrical circuit is designed to measure the current flowing through the atomic-scale contact with high sensitivity and low noise. Because the conductance of the junction can vary from several multiples of the conductance quantum down to values dominated by tunnelling, the measured currents may range from microamperes to picoamperes. To accommodate this wide range, the experiment employs a low-noise transimpedance amplifier (IV converter). The converter transforms the current through the junction into a proportional voltage signal using a feedback resistor whose value determines the measurement sensitivity. Selecting an appropriate gain is important: a high resistance increases sensitivity for tunnelling measurements but risks

saturating the amplifier during metallic contact, while a lower resistance trades sensitivity for a wider dynamic range [3].

The output of the IV converter is fed into a data acquisition (DAQ) unit, which digitises the signal with a resolution sufficient to resolve small variations in conductance. The DAQ is synchronised with the piezo controller so that each current measurement can be associated with a specific piezo voltage, enabling reconstruction of conductance traces as the junction is opened and closed. Additional series resistances may be incorporated in the circuit to stabilise the measurement or to protect the amplifier, although these are typically kept small to avoid influencing the intrinsic behaviour of the atomic contact.

Taken together, the piezo actuator, IV converter, and DAQ form a tightly coupled system. The actuator determines the mechanical evolution of the contact, while the electrical circuit records its response with high fidelity. This combination provides the temporal and spatial resolution needed to observe quantised conductance and to explore the transition between metallic and tunnelling transport [1, 7].

3.4 Software Functions

The measurement software plays a central role in operating the mechanically controlled break junction, as it coordinates piezo motion, data acquisition, and real-time visualisation of the junction's conductance. It provides a practical interface that allows the experimenter to monitor the contact as it evolves from a metallic connection to an atomic-scale constriction and finally into the tunnelling regime [4, 5].

One of the primary functions of the software is the live display of the measured current or conductance. As the piezo voltage is varied, the software continuously updates a plot of the conductance versus time or displacement. This direct visual feedback makes it possible to identify plateaus, sudden jumps, or instabilities in the junction. During the initial break-in phase, the live trace also helps the user determine whether the wire has fractured cleanly and whether the system is behaving as expected.

In addition to manual control, the software provides automated routines for opening and closing the junction. In such modes, the piezo voltage is swept back and forth at a controlled rate while the conductance is recorded at each step. Repeating this process over many cycles yields a large collection of conductance traces, which is essential for constructing histograms and identifying the most probable conductance values. The user can adjust the piezo speed, the total range of motion, and the sampling rate to tailor the measurement to different regimes of interest.

The software also includes a histogram mode that processes the accumulated conductance traces and compiles them into a probability distribution. Peaks in these histograms correspond to

conductance values at which the junction resides for extended periods, often aligning with integer multiples of the conductance quantum. This statistical approach provides a robust way to analyse conductance quantisation, even when individual traces display substantial variability due to atomic rearrangements [1, 7].

Finally, the software manages data logging. Each trace can be stored automatically with its corresponding piezo voltage, time stamp, and measurement parameters. This systematic archiving ensures that the recorded data can be analysed later, either to extract additional information or to compare different experimental conditions.

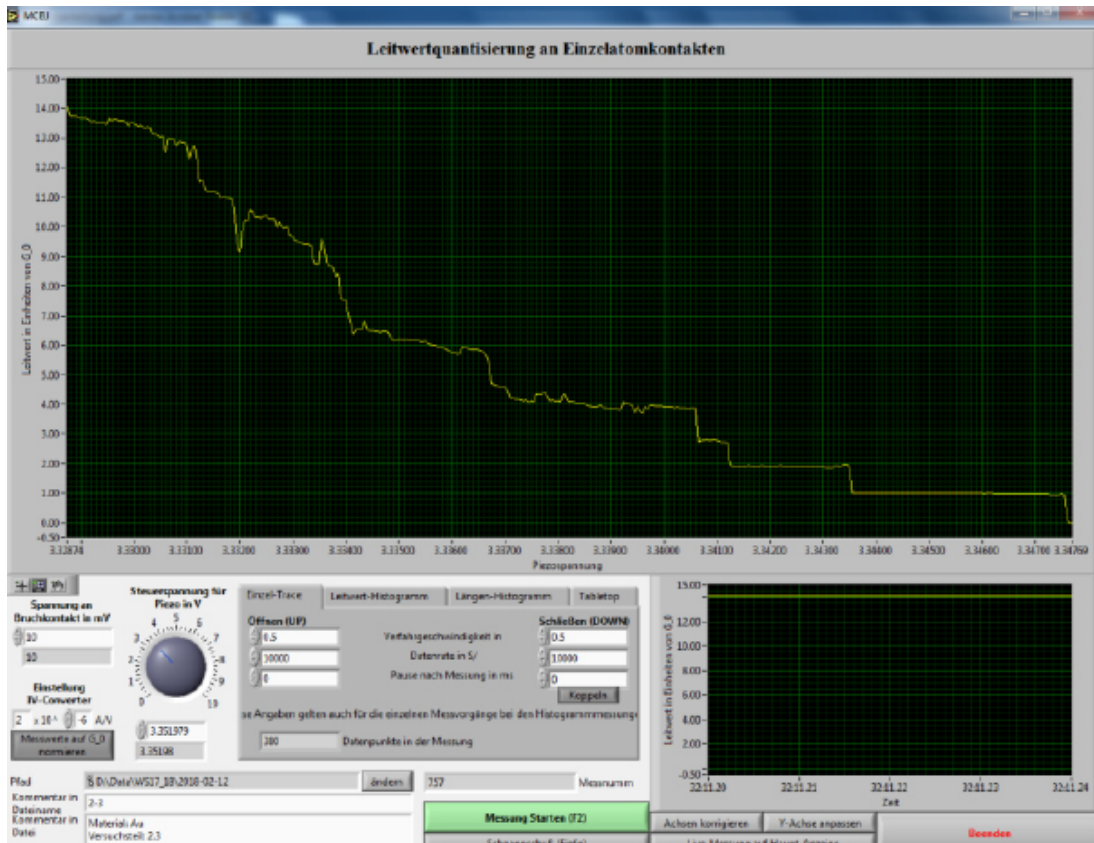


Figure 3.3: Example of the measurement software interface, showing a live conductance trace and options for automated opening–closing cycles and histogram generation [7].

4 Methodology

The following section outlines the procedure for performing the mechanically controlled break junction experiment. The steps are organised according to the sequence followed during the measurement and include essential safety considerations to ensure stable operation and reliable data [4, 5].

4.1 Establishing the Break Junction

The experiment begins with positioning the bending beam and the mounted wire so that the notch lies exactly above the central support. Before applying any voltage to the piezo, the wire is gently pre-stressed using the coarse mechanical adjustment. This slow manual winding increases the curvature of the substrate until the notched wire fractures at its weakest point. Care must be taken during this step to avoid sudden movements or mechanical shocks that might damage the clamps or misalign the electrodes.

Once the wire has broken, the electrode tips remain separated by only a few ångströms. The piezo actuator is then used to fine-tune the separation. Voltage is applied gradually, allowing the junction to be brought into either metallic contact or the tunnelling regime with precise control. During these adjustments, the system should remain free of vibrations, and external contact with the setup must be avoided, as even minor disturbances can alter the atomic configuration of the junction.

Proper grounding of all electronic components—including the IV converter and the DAQ unit—is essential at this stage to prevent noise pickup or electrical instabilities. Because the piezo operates at elevated voltages, its leads must also be handled carefully to avoid accidental discharge or damage to the driver electronics.

4.2 Recording Conductance Traces

With the junction established, the piezo voltage is swept back and forth to record conductance traces. In a typical run, the junction is opened until the conductance decays into the tunnelling regime and then closed again to reform an atomic contact. Repeating this cycle many

times reveals the characteristic plateaus and jumps resulting from atomic rearrangements in the constriction.

The piezo speed and the data acquisition rate must be chosen to balance temporal resolution with measurement stability. A slow actuation allows detailed observation of plateaus, while faster speeds are useful for collecting large datasets efficiently. Conductance curves generally exhibit near-flat regions close to integer multiples of the conductance quantum, which signal stable configurations of the atomic contact [1, 7].

To avoid artefacts in the measurement, the amplifier should not be driven into saturation, and unnecessary mechanical disturbances must be minimised during the recording window.

4.3 Calibration via Tunnelling

Before analysing the conductance steps quantitatively, the displacement of the piezo actuator must be calibrated. This is accomplished by placing the junction in the tunnelling regime, where the current decays exponentially with increasing electrode separation. By sweeping the piezo voltage slowly, the tunnelling current is recorded over several orders of magnitude.

Fitting the resulting data to an exponential dependence makes it possible to extract the decay constant κ , which reflects the effective barrier height. The derivative of the logarithmic current with respect to the piezo voltage provides the conversion factor between voltage and displacement. Combining this factor with the geometric reduction described earlier yields an estimate of the gear ratio for the specific setup [3].

During this calibration step, the amplifier operates at high sensitivity, so it is important to avoid electrical noise sources, unstable grounding, or static discharge.

4.4 Conductance Histograms

Once the calibration is complete and the measurement settings have been optimised, large sets of conductance traces can be acquired. Typically, between 500 and 1000 opening–closing curves are recorded to build a statistically meaningful histogram. Each individual trace contributes counts to bins based on the measured conductance values.

The resulting histogram reveals the most probable conductance levels at which the junction spends time. Prominent peaks often occur near integer multiples of G_0 , reflecting the stability of certain atomic configurations. Differences between opening and closing curves may appear, especially near the transition between metallic contact and tunnelling, due to hysteresis and atomic rearrangements as the junction is compressed and relaxed [8, 10].

Care must be taken to ensure that environmental disturbances do not bias the statistics. Vibrations, temperature fluctuations, or drifting electrical offsets can introduce systematic errors in the histogram.

5 Experimental Procedure

After completing the theoretical preparation and calibration considerations, the experiment was carried out using the mechanically controlled break junction (MCBJ) setup described in the instruction manual. The aim of the experimental procedure is to reproducibly form, open, and close an atomic-scale metallic contact while simultaneously measuring its electrical conductance. Particular care is required during the initial alignment and adjustment of the prepared sample, as this step determines the mechanical stability and controllability of the junction during subsequent measurements [4, 5].

5.1 Adjusting the Prepared Sample

Before mounting the prepared sample, all necessary equipment was switched on, including the control PC, the IV-converter, and the corresponding power supplies. The measurement software was then started, and a constant bias voltage of $U = 10$ mV was applied to the break junction. This low bias voltage ensures operation in the linear-response regime and avoids unnecessary heating of the contact.

Prior to reassembling the PVC-mounted sample holder onto the setup, the piezo actuator was fully wound down. This step is essential, as the maximum elongation range of the piezo crystal is limited to approximately $5\text{ }\mu\text{m}$. The PVC plate carrying the notched metallic wire was then carefully connected to the setup using the modified BNC connectors of the sample holder. At this stage, the closing of the electrical circuit could be directly observed in the software interface through the appearance of a finite conductance signal. The PVC plate was subsequently fixed in place using the four hexagonal screws provided.

To break the metallic junction for the first time, the piezo actuator was initially set to its maximum elongation. The junction was then opened by manually winding up the central push rod using the crank mechanism. In the initial phase, the crank could be rotated relatively quickly, as one full rotation corresponds to a vertical displacement of approximately $10\text{ }\mu\text{m}$. As the breaking point was approached, the adjustment was performed more cautiously in order to avoid damaging the sample or overshooting the breaking point.

Once the junction was successfully broken, it was verified that the contact could be opened and closed exclusively by varying the piezo elongation. If this was not possible, the coarse position of the piezo was readjusted using the crank. For stable operation, the breaking and closing point of the junction was positioned near the middle of the piezo elongation range. This configuration allows compensation for slow mechanical drifts during longer measurement sequences.

Finally, the sensitivity of the IV-converter was set to $2\text{ }\mu\text{A/V}$, which corresponds to the conductance range of interest for the subsequent measurements. With the junction properly adjusted and electrically stable, the setup was ready for the first conductance measurements.

5.2 First Conductance Measurements

After successfully adjusting the prepared sample, first conductance measurements were performed in order to characterise the basic behaviour of the atomic contact during opening and closing of the junction. These initial measurements serve as a qualitative test of the setup and allow suitable experimental parameters, such as piezo elongation speed and data acquisition rate, to be selected for all subsequent measurements [4, 5].

A constant bias voltage of $U = 10\text{ mV}$ was applied across the junction, ensuring operation in the linear-response regime. The conductance was recorded as a function of the piezo control voltage while the junction was slowly opened and closed.

Opening and Closing Traces

Representative conductance traces for both opening and closing of the junction are shown in Fig. 5.1 and Fig. 5.2, respectively. In both cases, the conductance exhibits a pronounced step-like behaviour, with plateaus occurring close to integer multiples of the conductance quantum $G_0 = 2e^2/h$.

During the opening of the junction, the conductance decreases in discrete steps as the atomic contact is gradually thinned and eventually broken. Several plateaus are visible up to conductance values of approximately $10G_0$, indicating the successive reduction of the number of transmitting channels. The presence of extended plateaus suggests that the junction opens sufficiently slowly to allow the formation of stable atomic configurations and, in some cases, short atomic chains [3].

When closing the junction, a similar step-like structure is observed, although the precise sequence and length of the plateaus differ from the opening trace. This hysteresis between opening and closing is expected and can be attributed to atomic rearrangements and plastic deformations of the contact, which depend on the mechanical history of the junction.

Chosen Measurement Settings

The experimental parameters were chosen such that a compromise between mechanical stability, temporal resolution, and data volume was achieved. From the data files used for the representative traces, the following settings were selected:

- Bias voltage at the junction: $U = 10 \text{ mV}$
- Sensitivity of the IV-converter: $2 \mu\text{A/V}$
- Piezo elongation speed:
 - Opening: 0.4 V/s
 - Closing: 0.9 V/s
- Data acquisition rate: 25 000 points per second

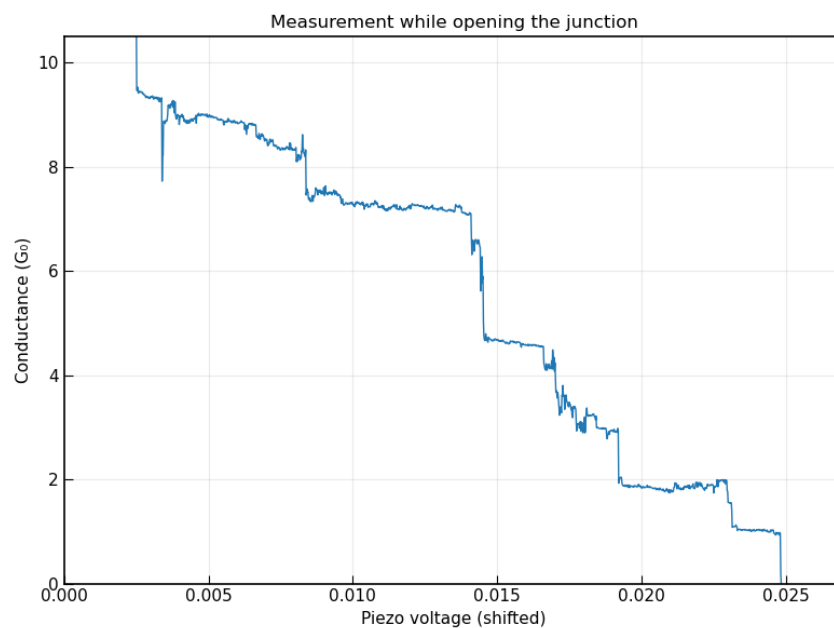


Figure 5.1: Representative conductance trace recorded while opening the gold junction. The conductance decreases in discrete steps, with plateaus close to integer multiples of G_0 .

These settings ensured that each conductance plateau consisted of several data points, while avoiding excessive data volumes and reducing the influence of mechanical vibrations.

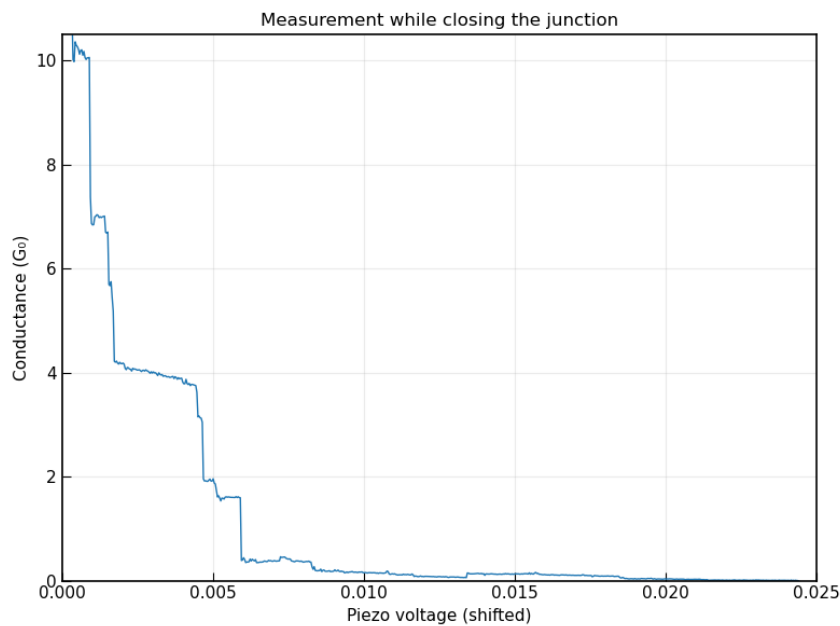


Figure 5.2: Representative conductance trace recorded while closing the gold junction. Step-like increases in conductance are observed, showing hysteresis compared to the opening trace.

Overall, the measured conductance traces are in good agreement with theoretical expectations for metallic atomic contacts. The appearance of plateaus near integer multiples of G_0 reflects the quantisation of conductance in terms of discrete transmission channels. Deviations from perfectly flat plateaus and exact integer values can be explained by thermal fluctuations, electronic noise, and variations in the atomic geometry of the contact. Small differences between the opening and closing traces are expected due to the stochastic nature of atomic rearrangements during deformation. These effects do not indicate experimental malfunction, but rather demonstrate the sensitivity of the MCBJ technique to atomic-scale structural changes.

5.3 Calibration Using the Tunnelling Regime

In the next step, the relation between the vertical displacement of the piezo actuator and the effective horizontal separation of the gold electrodes was calibrated. This calibration is necessary in order to convert the piezo control voltage into a physical length scale. For this purpose, the tunnelling regime of the junction was used, where the conductance depends exponentially on the electrode separation [3, 4].

When closing the junction slowly, a finite current can be measured even before the electrodes come into mechanical contact. In this regime, electron transport occurs via quantum tunnelling through the vacuum gap. As the distance between the electrodes decreases, the tunnelling

current increases exponentially until the contact finally jumps into the contact regime (jump-to-contact). This well-defined exponential dependence allows a precise calibration of the piezo displacement.

Measurement Procedure

For the calibration measurements, only a small conductance range close to zero, well below $1 G_0$, was considered in order to ensure operation in the pure tunnelling regime. Compared to the previous measurements, the data acquisition rate was increased to obtain a sufficiently large number of data points in the relevant region. All other experimental settings, such as bias voltage and IV-converter sensitivity, were kept unchanged.

A large number of individual closing traces (~ 100) were recorded. From these, only those traces were selected which exhibited a clear exponential increase of the conductance as the junction was closed. Traces showing sudden jumps, excessive noise, or deviations from exponential behaviour were excluded from the calibration analysis.

Fit Model and Extraction of the Calibration Factor

The tunnelling conductance was modelled using an exponential function of the form

$$G(V) = A \exp[B(V - V_0)], \quad (5.1)$$

where V is the piezo control voltage, V_0 denotes an offset voltage corresponding to an arbitrary reference position, and A and B are fit parameters. For the fits shown in this section, the offset voltage was fixed to

$$V_0 = 0.006389 \text{ V}.$$

A representative tunnelling curve together with the corresponding fit is shown in Fig. 5.3. From the fit parameter B , the calibration factor α , relating piezo voltage to electrode separation, was determined via

$$\alpha = -\frac{1}{B}. \quad (5.2)$$

The negative sign reflects the fact that the conductance increases as the electrode separation decreases.

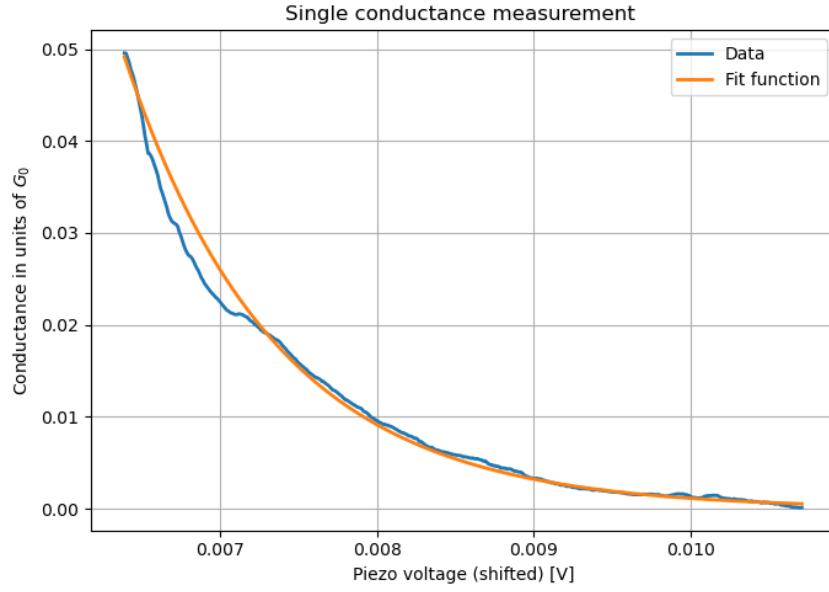


Figure 5.3: Representative tunnelling conductance trace recorded while closing the junction, together with an exponential fit used for calibration.

Statistical Analysis

In total, ten tunnelling curves fulfilled the strict selection criteria and were used for the final calibration. The resulting calibration factors were analysed statistically, yielding

$$\text{Mean } \alpha = (73.6 \pm 6.0) \text{ nm/V}, \quad (5.3)$$

$$\text{Standard deviation } \sigma_\alpha = 12.0 \text{ nm/V}, \quad (5.4)$$

$$\text{Median } \alpha = 70.2 \text{ nm/V}. \quad (5.5)$$

Here, the quoted uncertainty of the mean corresponds to the standard error of the mean (SEM). The relatively large spread reflects the sensitivity of the tunnelling current to small variations in atomic geometry and mechanical stability.

The experimentally obtained calibration factor is of the same order of magnitude as the estimate obtained during the preparation. Deviations between the calculated and measured values can be attributed to several factors, including uncertainties in the mechanical geometry of the setup, piezo hysteresis, and atomic-scale rearrangements of the electrode tips during repeated opening and closing cycles. This calibrated length scale is used in the following sections to analyse conductance histograms and distance-dependent phenomena quantitatively.

5.4 Generating Conductance Histograms

After establishing suitable measurement parameters and calibrating the piezo displacement, conductance histograms were generated in order to extract statistically significant information about the stability of atomic configurations in the gold junction. While individual conductance traces provide insight into single breaking events, histograms reveal which conductance values occur most frequently and are therefore energetically favoured during repeated opening cycles of the contact [3, 4].

All histogram measurements were performed by repeatedly opening the junction from a closed state. Prior to each measurement sequence, the contact was therefore fully closed to ensure comparable initial conditions. During histogram acquisition, both the evolving histogram and the individual conductance traces were continuously monitored. This allowed an immediate assessment of data quality and made it possible to identify representative single traces without extensive post-processing.

Measurement Strategy and Parameter Choice

The measurement settings determined in Section 5.2 were retained for histogram acquisition. In order to effectively average out statistical fluctuations arising from atomic rearrangements, a large number of individual opening traces was recorded. For each histogram, more than 500 single conductance measurements were included, providing sufficient statistical weight while keeping the total data volume and measurement time within reasonable limits.

The conductance range was restricted to values below approximately $10 G_0$, since higher conductance values as well as the completely opened contact give rise to artificially high counts that do not reflect intrinsic properties of the atomic junction. Similarly, the upper boundary of the measurement range was excluded to avoid distortions caused by saturation effects.

The bin width of the histogram bars was chosen such that the entire conductance range was covered uniformly. The binning was fine enough to resolve individual conductance plateaus, yet coarse enough to suppress small fluctuations and electronic noise. This choice ensures that the resolution of the data acquisition system does not limit the physical interpretation of the histogram.

Conductance Histograms and Interpretation

The resulting conductance histograms for opening and closing measurements are shown in Fig. 5.4. Pronounced peaks are observed near integer multiples of the conductance quantum G_0 , indicating preferred atomic configurations with a discrete number of transmission channels.

These peaks arise from the repeated formation of similar atomic structures during successive breaking events.

Differences between the opening and closing histograms can be attributed to the different atomic pathways followed during deformation. While opening the junction tends to favour the formation of atomic chains and well-defined plateaus, closing the contact often involves more abrupt structural rearrangements, leading to broader or suppressed peaks.

Overall, the histogram analysis confirms the quantised nature of conductance in atomic-scale gold contacts and demonstrates the importance of statistical averaging for identifying robust physical features beyond individual breaking events.

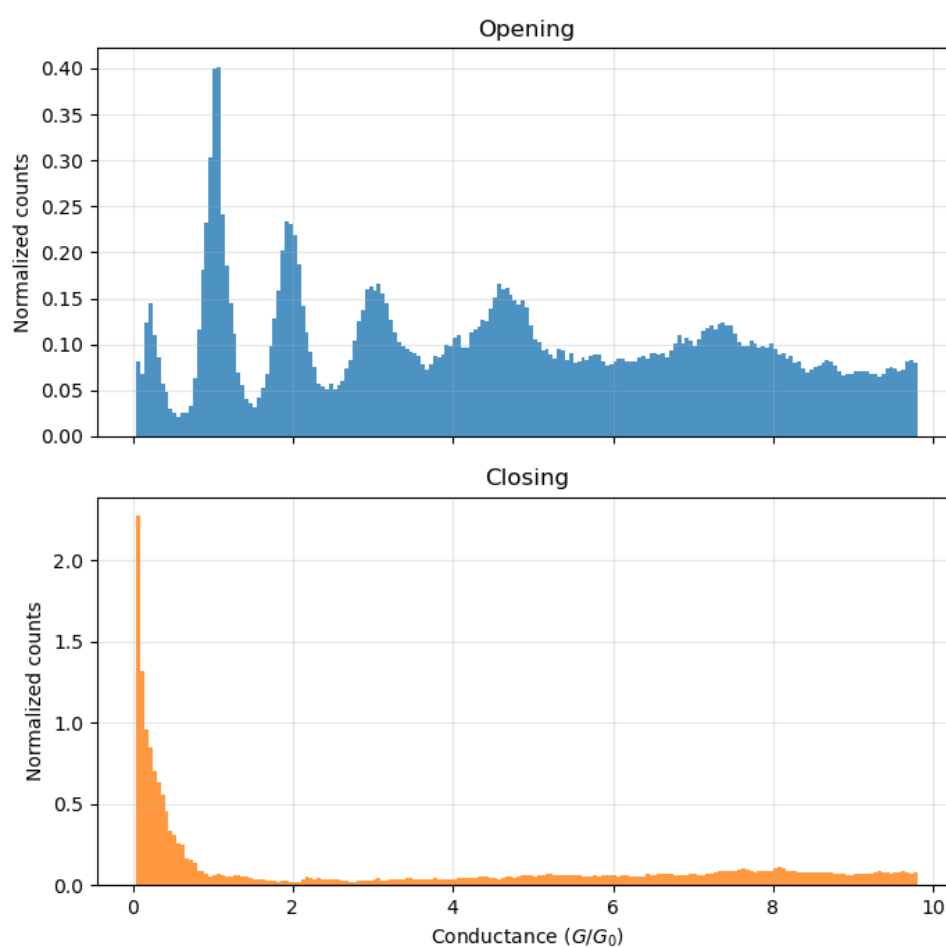


Figure 5.4: Conductance histograms for opening (top) and closing (bottom) measurements. Peaks near integer multiples of G_0 indicate preferred atomic configurations of the gold junction.

Averaged Conductance Traces

In addition to the histograms, averaged conductance traces were constructed separately for opening and closing cycles. Figures 5.5 and 5.6 show the mean conductance as a function of the shifted piezo voltage, together with the corresponding spread of the data.

The averaged traces reproduce the step-like decrease in conductance observed in individual measurements, while significantly reducing random fluctuations. Differences between opening and closing traces are clearly visible, reflecting hysteresis effects caused by atomic-scale rearrangements and plastic deformation of the contact.

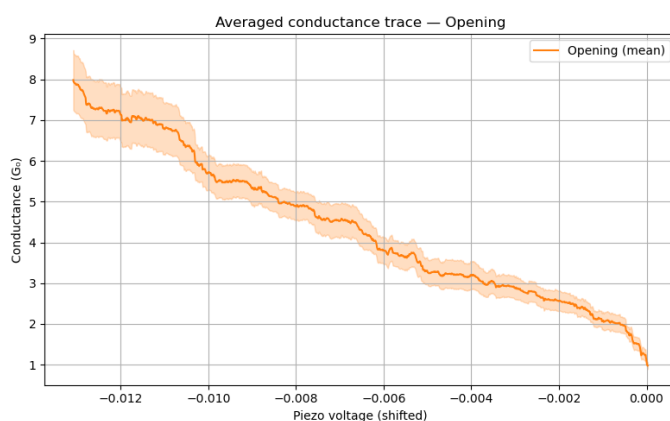


Figure 5.5: Averaged conductance trace for repeated opening cycles of the junction. Step-like features remain clearly visible after averaging.

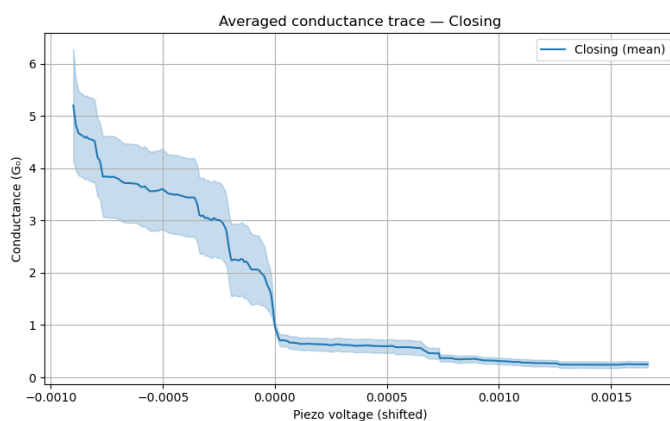


Figure 5.6: Averaged conductance trace for repeated closing cycles of the junction. The shaded region indicates the spread of the individual measurements around the mean.

5.5 Shell Effect

In a next step, conductance measurements were analysed with the aim of investigating shell effects at higher conductance values. In contrast to the low-conductance regime, where well-defined plateaus near integer multiples of G_0 are observed, the conductance at higher values is dominated by an increasing number of partially transmitting channels. As a result, individual plateaus become less pronounced and conductance histograms exhibit broadened features [3, 7].

According to the instruction manual, this investigation is ideally carried out over a conductance range extending up to $100 G_0$. In the present experiment, however, the maximum conductance reached was limited to approximately $G_{\max} \approx 14 G_0$. Consequently, the analysis of shell effects was restricted to the experimentally accessible range, which is explicitly taken into account in the following discussion.

Physical Origin of Shell Effects

Enhanced stability at certain conductance values is commonly attributed to two complementary mechanisms:

- **Electronic shell effect:** Certain contact cross-sections correspond to energetically favourable filling of electronic states or transmission modes. When such electronic shells are effectively closed, the total energy of the junction is reduced, leading to an increased lifetime of the corresponding configuration and to local maxima in conductance histograms.
- **Geometric shell effect:** Enhanced stability can also arise from completed atomic layers or facets in the contact geometry. Such compact atomic arrangements are mechanically more robust and therefore persist longer during junction deformation, again leading to increased histogram counts at specific conductance values.

At room temperature and for large contact cross-sections, both effects are expected to be weak and strongly broadened due to thermal motion, channel mixing, and atomic rearrangements.

Histogram Analysis and Representative Traces

Conductance histograms for opening and closing cycles were generated using all available traces in the accessible conductance range. Figure 5.7 shows the histogram for opening traces, together with a smoothed curve used to identify local maxima. While the histogram is largely smooth, several weak and irregular maxima can be identified, suggesting that certain configurations occur more frequently than others.

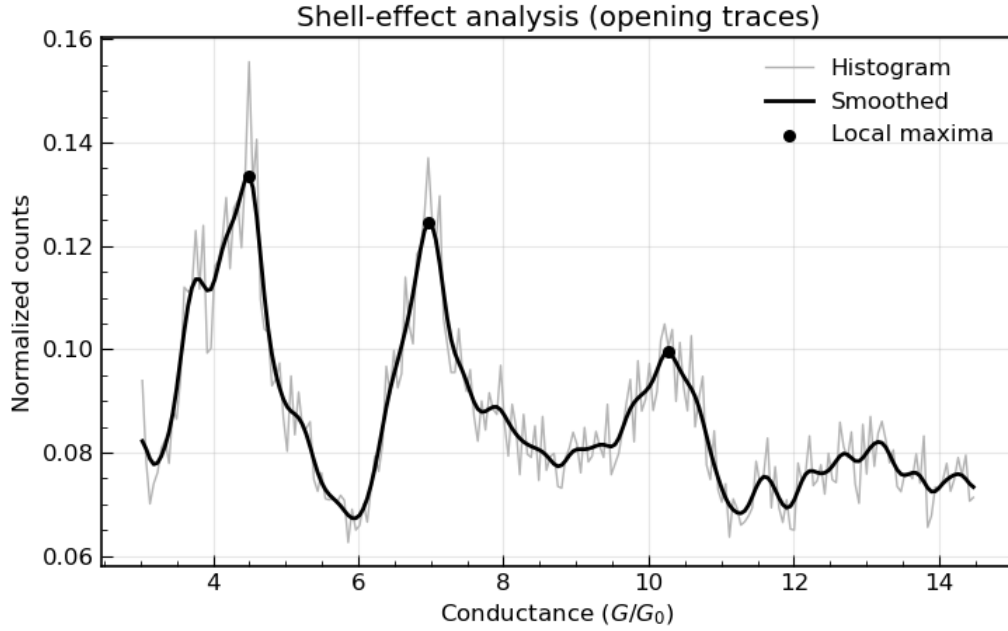


Figure 5.7: Conductance histogram for opening traces in the high-conductance regime accessible in this experiment. The smoothed curve highlights weak local maxima, indicating preferred configurations despite strong broadening.

Representative single conductance traces corresponding to high conductance values exhibit a continuous and irregular evolution rather than sharply defined plateaus. This behaviour is consistent with transport through many interacting transmission channels and reflects the absence of well-separated conductance modes in this regime.

Experimental Limitations and Discussion

The limited conductance range represents the main restriction of the present shell-effect analysis. Since the measurements did not extend to $100 G_0$, it was not possible to generate histograms or representative single traces covering the full high-conductance regime specified in the instruction manual. As a result, a clear identification or separation of electronic and geometric shell effects was not feasible.

Nevertheless, the experimental observations are consistent with theoretical expectations. Shell-related features at higher conductance values are expected to be weak, irregular, and strongly broadened, particularly under room-temperature conditions and for limited datasets. Differences between opening and closing measurements are generally expected due to hysteresis effects and different atomic rearrangement pathways during deformation. However, in the present analysis only opening traces were available in the high-conductance regime, preventing a direct comparison between opening and closing behaviour.

In conclusion, the shell-effect analysis illustrates the gradual disappearance of well-defined conductance features with increasing contact size. Although the full analysis outlined in the instruction manual could not be completed due to experimental constraints, the results provide qualitative insight into the transition from quantised conductance at low channel numbers to a regime dominated by complex multi-channel transport. The present measurements highlight both the physical limitations and the experimental sensitivity of the MCBJ technique when investigating shell effects at high conductance.

5.6 Recording Length Histograms

This part of the experiment focuses on the formation and stability of single-atom contacts. In mechanically controlled break junction experiments, the last conductance plateau before the contact ruptures is typically associated with a single gold atom bridging the two electrodes. The length of this final plateau provides information about the mechanical stability and elongation behaviour of such single-atom contacts.

To quantify this behaviour, a length histogram was constructed using the built-in recording feature of the acquisition software. For each opening trace, only the final conductance plateau close to $1 G_0$ was considered. In order to reliably identify well-defined single-atom plateaus while suppressing transient crossings and noise, a strict selection window of $\pm 20\%$ around the measured mean value of $1 G_0$ was applied, corresponding to

$$0.8 G_0 \leq G \leq 1.2 G_0.$$

For each valid trace, exactly one plateau length was extracted and added to the histogram.

Length Histogram

Figure 5.8 shows the resulting length histogram for the last conductance plateau. The horizontal axis corresponds to the plateau length expressed in piezo units, while the vertical axis shows the probability density. The histogram exhibits a pronounced accumulation at very short plateau lengths, with only a small number of longer-lived plateaus.

Only a limited number of valid single-atom plateaus were identified within the chosen selection window. This indicates that most junctions either pass quickly through the $1 G_0$ conductance value or rupture before a stable single-atom configuration can be maintained.

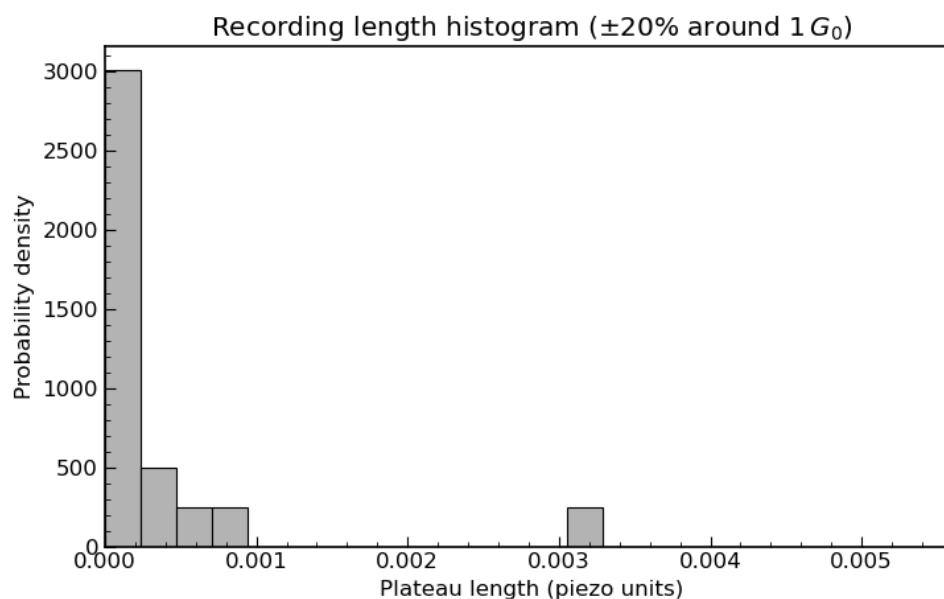


Figure 5.8: Length histogram of the last conductance plateau around $1 G_0$ using a $\pm 20\%$ selection window. Only well-defined single-atom plateaus contribute to the histogram.

Experimental Limitations and Discussion

The small number of recorded single-atom plateau lengths is not unexpected. The formation of a true single-atom gold contact requires a very specific atomic arrangement, which is statistically rare under room-temperature conditions. Thermal motion leads to rapid atomic rearrangements, significantly reducing the lifetime of stable single-atom configurations. As a result, long-lived plateaus near $1 G_0$ occur only infrequently.

The strict $\pm 20\%$ selection criterion further reduces the number of accepted plateaus. While a wider conductance window would increase the number of counts, it would also include transient crossings and noisy segments, thereby degrading the physical significance of the histogram. The chosen criterion therefore represents a deliberate trade-off between statistical yield and reliability, in accordance with the recommendations of the instruction manual.

In addition, the total number of available opening traces in the present experiment was significantly smaller than the approximately 10^4 measurements suggested in the manual. Given this limitation, the identification of only a limited number of well-defined single-atom plateaus is reasonable and does not indicate experimental malfunction.

Using the calibration factor obtained from the tunnelling analysis, the extracted plateau lengths were converted from piezo units into physical length scales. However, due to the limited number of well-defined single-atom plateaus identified within the strict $\pm 20\%$ selection window, a statistically reliable determination of a mean plateau length was not possible.

This limitation is a direct consequence of insufficient statistics. Only a small fraction of all opening traces form stable single-atom contacts that persist long enough to be recorded as plateaus near $1 G_0$. As a result, the length histogram is dominated by short-lived events (

$$\langle L \rangle \approx 1.0 \times 10^{-10} \text{ nm.}$$

), while longer plateaus occur too rarely to allow a meaningful quantitative average.

The absence of a well-defined mean length therefore reflects the experimental conditions. In particular, room-temperature measurements, a limited total number of traces, and the strict selection criterion significantly reduce the number of statistically relevant single-atom plateau lengths.

Consequently, the length histogram is dominated by plateau lengths close to the detection limit, leading to a strongly reduced mean value. Longer plateaus, which would correspond more directly to atomic-scale elongations or short atomic chains, are statistically rare in room-temperature measurements with limited statistics [3, 7].

5.7 Influence of Voltage on the Break Junction

In this part of the experiment, the influence of the applied bias voltage on the behaviour of the break junction was investigated. While the previous measurements were performed at a fixed low bias voltage, increasing the voltage across the junction can significantly affect the atomic configuration and stability of the contact. At higher voltages, enhanced Joule heating, increased electromigration, and voltage-induced forces may alter the conductance characteristics and the lifetime of atomic configurations [3, 7].

According to the instruction manual, this study should ideally be carried out using multiple conductance histograms recorded at different applied voltages, each containing a large number of individual opening traces. In the present experiment, measurements were performed for a set of bias voltages ranging from 5 mV to 400 mV. Due to time and data limitations, only one opening histogram was available for each voltage setting. This restriction is explicitly taken into account in the following qualitative discussion.

Comparison of Conductance Histograms

Figure 5.9 shows conductance histograms recorded for different applied bias voltages, presented in an offset representation to facilitate direct comparison. All histograms are normalised and plotted over the same conductance range, with the position of $1 G_0$ indicated by a dashed line.

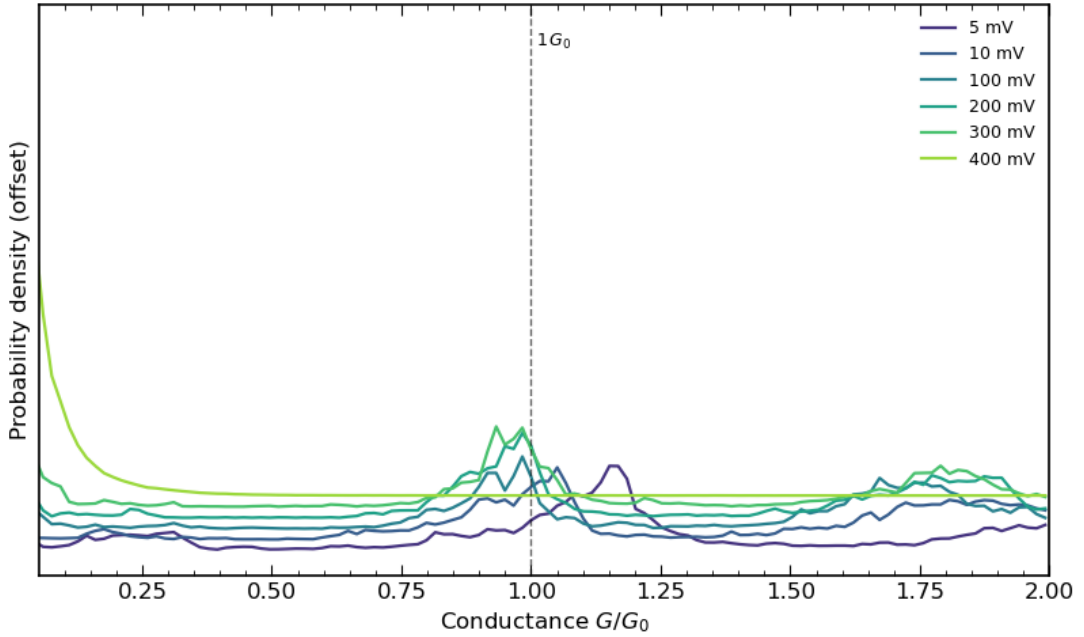


Figure 5.9: Conductance histograms of opening traces recorded at different applied bias voltages. For clarity, the histograms are vertically offset. The dashed line indicates the conductance quantum G_0 .

At low bias voltages (5 mV and 10 mV), a pronounced accumulation of counts near $1 G_0$ is observed, reflecting the relatively high stability of single-atom contacts under low electrical stress. As the applied voltage is increased, the peak near $1 G_0$ becomes progressively broader and less pronounced. At higher voltages, additional conductance fluctuations and a redistribution of histogram weight are visible.

Discussion

The observed evolution of the conductance histograms with increasing bias voltage is consistent with physical expectations. Higher applied voltages lead to increased power dissipation in the atomic contact, resulting in local heating and enhanced atomic mobility [3]. These effects reduce the lifetime of well-defined atomic configurations and promote faster structural rearrangements, which in turn smear out the conductance plateaus.

In addition, electromigration forces acting on the atoms become more significant at elevated voltages, further destabilising narrow constrictions [3, 7]. As a consequence, the probability of forming long-lived single-atom contacts decreases, and the conductance histograms lose their sharp features.

It should be emphasised that, due to the limited number of traces available for each voltage setting, the present results allow only a qualitative comparison. A statistically robust analy-

sis would require a substantially larger number of opening traces for each applied voltage, as suggested in the instruction manual. Nevertheless, the systematic broadening and weakening of conductance features with increasing bias voltage clearly demonstrate the influence of the applied voltage on the mechanical and electronic stability of the break junction. The applied bias voltage has a noticeable impact on the behaviour of the break junction. Low voltages favour the formation of stable, well-defined conductance plateaus, while higher voltages lead to increased fluctuations and reduced stability of atomic configurations. Although the present analysis is limited by insufficient statistics, the observed trends are in good qualitative agreement with theoretical expectations and illustrate the role of voltage-induced effects in atomic-scale contacts.

5.8 Tabletop Experiment

The quantisation of conductance can also be demonstrated using a simplified “tabletop” setup, in which the electrical contact is formed by gently crossing two short pieces of wire. In contrast to the mechanically controlled break junction (MCBJ), the opening and closing of the contact is not driven by a piezo actuator but occurs in an uncontrolled manner through small vibrations and manual motion. Repeated brief losses of contact lead to gradual plastic deformation and thinning of the constriction, such that the junction can transiently reach an atomic-scale contact shortly before rupture. In this regime, conductance plateaus near integer multiples of G_0 are expected due to transport through discrete conduction channels [3, 4].

Experimental Setup

Two short wire pieces (~ 5 cm each) were mounted and crossed to form a point contact. Clips, cables, and adapters provided in the lab were used to connect the wires electrically to the measurement electronics. The wires were mechanically stabilised using tape in order to reduce large-scale motion while still allowing small vibrations to intermittently break and re-establish the contact. Gold was chosen as the primary material because it is mechanically ductile and exhibits pronounced conductance quantisation in atomic contacts.

Selection Criteria

Because the contact formation and rupture are not controlled, the acquired traces contain a wide range of behaviours, including very short events, long drifting contacts, and segments without meaningful plateau structure. To obtain a histogram with good statistics while rejecting clearly unusable traces, selection criteria were applied in the software:

- **Trigger-point logic:** a valid trace was required to pass through the trigger thresholds in the intended order (contact formation followed by rupture), ensuring that only full opening events were recorded.
- **Minimum/maximum curve length:** traces that were too short (dominated by noise or single spikes) or too long (strong drift, sticking contacts, or incomplete rupture) were excluded.

These criteria were chosen to keep as many traces as possible while discarding measurements that do not represent atomic-scale thinning. Additional useful criteria that would further improve selection, but are not directly adjustable in the tabletop configuration, would include filtering based on plateau flatness (derivative threshold), exclusion of traces with saturation/overload, and constraints on the time spent in the tunnelling regime (to reject large contact-area drifts).

Histogram with Reliable Statistics

A conductance histogram for gold was generated from the accepted traces. Due to the simplicity of the setup, a large number of data points could be collected within a short time, providing robust statistics. Figure 5.10 shows the resulting histogram (gold), with a total of 74,423 samples included. Peaks are visible near integer multiples of G_0 , consistent with the formation of preferred atomic configurations during repeated thinning and rupture events [3].

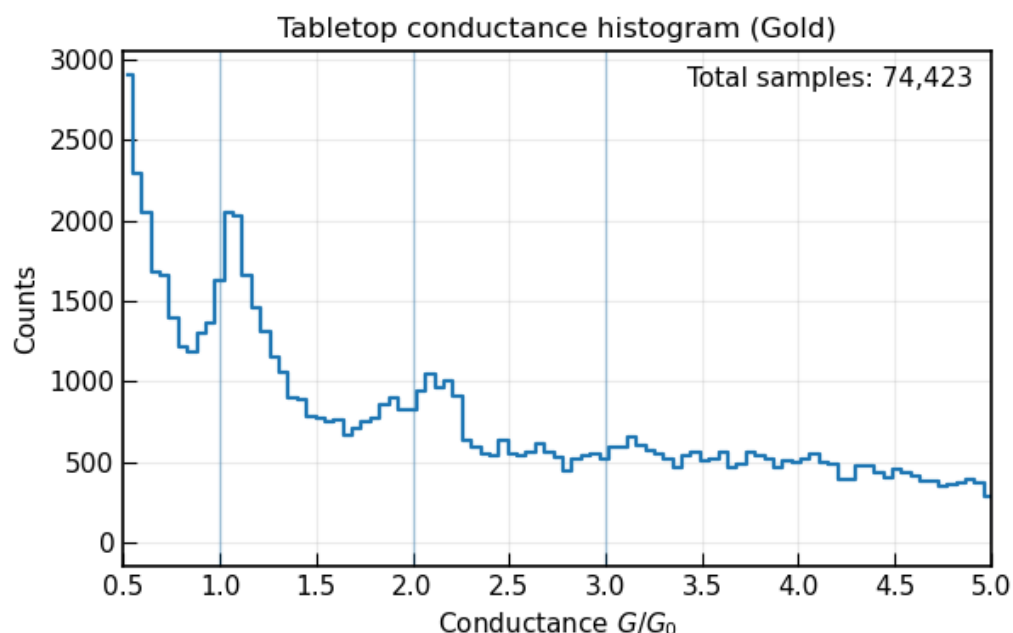


Figure 5.10: Tabletop conductance histogram for gold. The large number of samples yields reliable statistics, and enhanced counts near integer multiples of G_0 indicate preferred atomic configurations during repeated contact rupture.

Comparison with the MCBJ Histogram

To compare the tabletop approach with the controlled MCBJ measurement, the conductance histogram obtained in the tabletop experiment was plotted together with the corresponding MCBJ histogram over the range up to $10 G_0$ (Fig. 5.11). While both measurements show enhanced probability near characteristic conductance values, the tabletop histogram generally exhibits broader features due to the less controlled mechanical conditions. In the MCBJ setup, the piezo-driven motion provides smoother and more reproducible thinning trajectories, typically leading to sharper and more stable plateaus. In the tabletop setup, mechanical noise, uncontrolled contact area changes, and varying deformation pathways cause stronger broadening and increased background contributions.

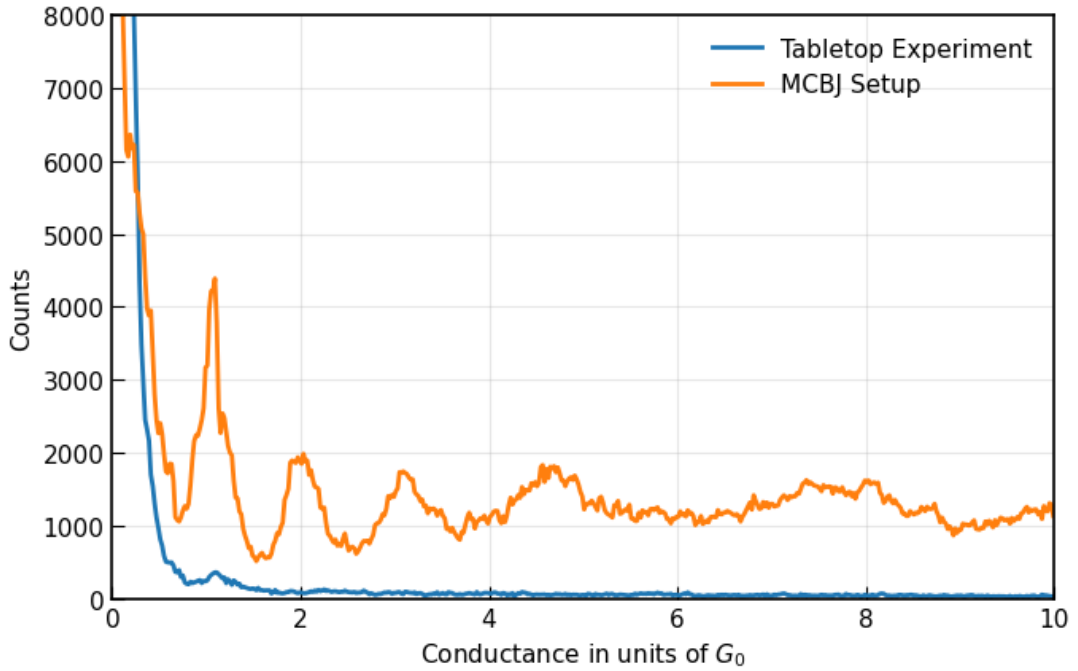


Figure 5.11: Comparison of conductance histograms obtained in the tabletop experiment and in the MCBJ setup (gold), shown in absolute counts over the range up to $10 G_0$. The tabletop histogram exhibits broader features due to uncontrolled mechanical motion, while the MCBJ measurement yields more reproducible plateau structures.

According to the instruction manual, a second conductance histogram should be recorded for a different material and compared to the gold measurement. In the present experiment, this additional measurement could not be completed due to time constraints and insufficient data acquisition for a second material.

As a result, the analysis in this section is restricted to the gold tabletop experiment, which serves as the reference case. A meaningful comparison between different materials would require

a sufficiently large number of traces for each material in order to obtain reliable statistics and to distinguish material-specific effects on the conductance histogram. Such a comparison is therefore beyond the scope of the present report.

In conclusion, the tabletop experiment provides a simple and robust demonstration of conductance quantisation without requiring a mechanically controlled junction. Although the opening and closing of the contact are less controlled than in the MCBJ setup, reliable statistics can be achieved quickly, and conductance maxima near integer multiples of G_0 are still observable. The comparison with the MCBJ highlights the trade-off between experimental simplicity and mechanical reproducibility: the tabletop setup is fast and accessible, while the MCBJ provides finer control and typically cleaner plateau structure.

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